User Stories - Intelligent Bus Driver Guidance System

ISYS3413 / ISYS3475 / ISYS1118 - Software Engineering Fundamentals - Assignment 4 (Team G17). The following eight user stories follow the INVEST principle and the 'As <role>, I want <goal>, so that <benefit>' template introduced in Week 8. Each story is accompanied by three acceptance criteria written in the Given-When-Then style so that the conditions of satisfaction are objectively testable.

User Story 1	
Role	As a bus driver,
Goal	I want to see turn-by-turn route guidance on my touchscreen,
Benefit	so that I can follow the assigned route accurately without taking my eyes off the road for long.
Acceptance Criteria	1. Given a trip has been started, when the engine is on, then the next-direction prompt appears on the touchscreen within 2 seconds, showing the distance to the next manoeuvre. 2. Given the bus is within 200 metres of a turn, when the GPS confirms the location, then the on-screen indicator flashes and a single audio cue is played. 3. Given GPS signal is lost for more than 10 seconds, when the loss is detected, then the system displays 'GPS lost - using last known position' and continues to show the most recent instruction until signal returns.

User Story 2	
Role	As a bus driver,
Goal	I want to receive a stop-approach alert before each scheduled stop,
Benefit	so that I can decelerate in time and stop safely without exceeding the comfort limits for passengers.
Acceptance Criteria	1. Given the bus is 150 metres from a scheduled stop, when the system computes the distance, then an audio chime plays once and the stop name is displayed on the screen for at least 5 seconds. 2. Given the bus is within 30 metres of the stop, when the GPS confirms proximity, then the screen displays 'Approaching stop - prepare to decelerate' in a high-contrast banner. 3. Given the next stop is a request stop with no passenger requests, when the alert would normally fire, then the alert is suppressed and the driver is informed that the stop may be bypassed.

User Story 3	
Role	As a bus driver,
Goal	I want to be informed when I have deviated from the planned route,
Benefit	so that I can return to the correct path quickly and reduce service delay.
Acceptance Criteria	1. Given the planned route is loaded, when the GPS position is more than 50 metres off the route for over 10 seconds, then the system flags a 'Route deviation' state. 2. Given a route deviation has been flagged, when the state changes, then a visual banner and one audio alert ('Off route - recalculating') are displayed within 3 seconds. 3. Given a route deviation event has been raised, when it is detected, then the timestamp, GPS coordinates and the active driverID are appended to the deviation log file.

User Story 4	
Role	As a bus driver,
Goal	I want to receive an automatic detour suggestion when my route is blocked,
Benefit	so that I do not have to navigate manually and the service continues with minimum disruption.
Acceptance Criteria	1. Given the traffic management system has notified the bus of a road closure on the planned route, when the notification is received, then an alternative route is computed and shown within 15 seconds. 2. Given a detour has been computed, when it is displayed, then the new segment is highlighted in blue on the map and the estimated extra travel time is shown next to the 'Accept' button. 3. Given the detour is presented, when the driver taps 'Accept' or 'Decline', then the choice and the timestamp are written to the trip log for later dispatch review.

User Story 5	
Role	As a bus driver,
Goal	I want to report a road hazard from the touchscreen in a few taps,
Benefit	so that dispatch and the following buses are warned promptly and can react.

User Story 5	
Acceptance Criteria	1. Given any active screen is visible, when the driver wants to report a hazard, then the 'Report Hazard' button is reachable within two taps from that screen. 2. Given the 'Report Hazard' screen is open, when a hazard type (pothole, accident, debris, flood, other) is selected, then the current GPS location is auto-populated and the report is sent on one further tap. 3. Given a hazard report has been submitted, when the dispatch system has acknowledged it, then the screen displays 'Hazard reported - dispatch notified' within 5 seconds and a log line containing the timestamp, location, type and driverID is appended to the hazard log file.

User Story 6	
Role	As a bus driver,
Goal	I want to report a delay using a voice command,
Benefit	so that I can keep both hands on the steering wheel while informing dispatch.
Acceptance Criteria	1. Given the driver presses the dedicated voice button, when the press is detected, then listening mode is activated within 1 second and a microphone icon is shown on the screen. 2. Given the driver says 'Report delay <reason>', when the system recognises the phrase, then it asks for verbal or on-screen confirmation before forwarding anything to dispatch. 3. Given the driver confirms the spoken report, when the confirmation is captured, then the delay event including the reason and estimated extra duration is forwarded to the dispatch system and stored locally.

User Story 7	
Role	As a bus driver,
Goal	I want to confirm arrival at each scheduled stop,
Benefit	so that the system holds an accurate record of stop times for service-reliability reporting.
Acceptance Criteria	1. Given the bus is stationary within 20 metres of a scheduled stop, when the bus has stayed stationary for more than 5 seconds, then a 'Confirm Arrival' button becomes visible on the screen.

User Story 7	
	2. Given the driver taps 'Confirm Arrival', when the tap is registered, then the timestamp, stop ID and dwell time are appended to the trip-record TXT file. 3. Given the driver does not tap 'Confirm Arrival' before leaving the stop area, when the GPS exit is detected, then the system auto-confirms the arrival using the entry/exit timestamps so that no stop is missed in the log.

User Story 8	
Role	As a dispatch operator,
Goal	I want to view the real-time status of every active bus,
Benefit	so that I can monitor service performance and respond to incidents as soon as they arise.
Acceptance Criteria	1. Given the dispatch dashboard is open, when 30 seconds have elapsed since the last refresh, then every bus's location, ETA-to-next-stop, current driver and on-time/delayed status are refreshed automatically. 2. Given a bus has an active hazard report or route deviation, when the dashboard is rendered, then that bus appears at the top of the list with a coloured indicator next to its busID. 3. Given the operator selects a bus, when the detail view loads, then a 24-hour trip history sourced from the trip TXT log files is displayed within 5 seconds.

Total: 8 user stories